

```
%%This lab is performed by Aditi Jain(UI24EC02)
```

```
%VARIABLE CREATION
```

```
x=pi;  
y=sin(10);  
z=3.14;
```

```
%MATRIX
```

```
a=[3 4 5 ;6 7 8]
```

```
a = 2×3  
    3    4    5  
    6    7    8
```

```
b=1:10
```

```
b = 1×10  
    1     2     3     4     5     6     7     8     9    10
```

```
c=1:0.5:4
```

```
c = 1×7  
    1.0000    1.5000    2.0000    2.5000    3.0000    3.5000    4.0000
```

```
d=linspace(1,10,5)
```

```
d = 1×5  
    1.0000    3.2500    5.5000    7.7500   10.0000
```

```
e=rand(2)
```

```
e = 2×2  
    0.9572    0.8003  
    0.4854    0.1419
```

```
f=zeros(3,3)
```

```
f = 3×3  
    0     0     0  
    0     0     0  
    0     0     0
```

```
g=ones(4)
```

```
g = 4×4  
    1     1     1     1  
    1     1     1     1  
    1     1     1     1  
    1     1     1     1
```

```
display(a)
```

```
a = 2×3  
    3     4     5  
    6     7     8
```

```
display(b)
```

```
b = 1×10
    1     2     3     4     5     6     7     8     9    10
```

```
display(c)
```

```
c = 1×7
    1.0000    1.5000    2.0000    2.5000    3.0000    3.5000    4.0000
```

```
display(d)
```

```
d = 1×5
    1.0000    3.2500    5.5000    7.7500   10.0000
```

```
display(e)
```

```
e = 2×2
    0.9572    0.8003
    0.4854    0.1419
```

```
display(f)
```

```
f = 3×3
    0     0     0
    0     0     0
    0     0     0
```

```
display(a(end,2))
```

```
7
```

```
display(a(1:2,end))
```

```
5
8
```

```
a(2)=121
```

```
a = 2×3
    3     4     5
   121    7     8
```

```
x1=a+1;
display(x1)
```

```
x1 = 2×3
    4     5     6
   122    8     9
```

```
y1=a.*a
```

```
y1 = 2×3
    9     16    25
  14641    49    64
```

```
display(y1)
```

```
y1 = 2×3
    9     16    25
  14641    49    64
```

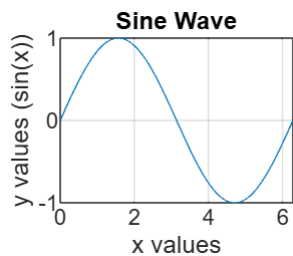
```
z1=a.*a
```

```
z1 = 2×3
      9      16      25
    14641    49      64
```

```
%PLOT
```

```
x5 = linspace(0, 2*pi, 1000);
```

```
y5 = sin(x5);
plot(x5, y5);
xlabel('x values');
ylabel('y values (sin(x))');
title('Sine Wave');
grid on;
```



```
%%CONDITIONAL
```

```
t=a>21;
display(t)
```

```
t = 2x3 logical array
    0     0     0
    1     0     0
```

```
%PROGRAMING
```

```
if x > 0.5
q = 3
else
q = 4
end
```

```
q =
3
```

```
display(q)
```

```
q =
3
```

```
for l = 1:3
disp(l)
```

end

1

2

3